MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 833A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

See figure 1

Weight: 1-pound (453.6 grams) nominal

Mounting position: Vertical; filament end either up or down.

Horizontal; with plane of electrodes vertical (on edge).

ABSOLUTE RATINGS: C Telegraphy

Parameter:	F	Ef	Eb	Ec	Ib	Ic	Pp	Pi
Unit:	MHz	V	kVdc	Vdc	mAdc	mAdc	W	W
Maximum:	30	10.5	3.0	-500	500	100	300	1,250
Minimum:	---	9.5	---	---	---	---	---	---
TEST CONDITIONS:	---	10.0 Vac	2.5	-10	---	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	9.4	10.6	Aac
1256	Electrode current (anode)	-	Eb = 2.5 kVdc; Ec = -50 Vdc	0.65	II	Ib	60	140	mAdc
1266	Total grid current	3	Eb = 2 kVdc; Ec/Ib = 150 mAdc; t = 300 seconds	0.65	II	Ic	0	-15	µAdc
	Quality conformance inspection, part 2								
1036	Bump	-	Angle = 25°	---	---	---	---	---	---
1316	Amplification factor	-	Eb/Ib = 200 mAdc	6.5	S3	Mu	31.5	38.5	---
(B)1236	Power oscillation	-	Ib = 450 mAdc; Ic = 80 to 120 mAdc; Rg = 5,000 ohms; Eb = 4.0 kVdc; t = 300 seconds; F = 25 to 30 MHz	6.5	S3	Po	1,000	---	W (useful)
(B)1231	Emission	3	Ib = 450 mAdc; Ic = 80 to 120 mAdc; Rg = 5,000 ohms; Eb = 4.0 kVdc; t = 300 seconds; F = 25 to 30 MHz Ef/Po = -10%	0.65	II	Ef	---	9.0	Vdc
1331	Direct-interelectrode capacitance			---	---	Cgp	5.5	7.1	pF
						Cin	10.1	14.5	pF
						Cout	6.4	10.6	pF
	Quality conformance inspection, part 3								
---	Life test	-	Group D; Ib = 410 mAdc; Ic = 80 to 120 mAdc; Rg = 5,000 ohms; Eb = 3.0 kVdc; Pp = 300 W; F = 30 MHz; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Emission	-		---	---	Ef	---	9.5	Vdc
1031	Low-frequency vibration	2	Eb = 250 Vdc; Ec/Ib = 5 mAdc; Rp = 2,000 ohms	---	---	Ep	---	1,000	mVac

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NOTES:

1. Forced air cooling required. An airflow of approximately 40 cfm from a 2-inch diameter nozzle directed on the bulb between grid and anode seals should be adequate.

Ⓑ 2. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection (QCI), part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

3. This test to be one of the first tests performed at the conclusion of the holding period in the following order:

   (1) Total grid current; (2) Emission.

Custodians:
Army - EL
Navy - EC
Air Force - 85

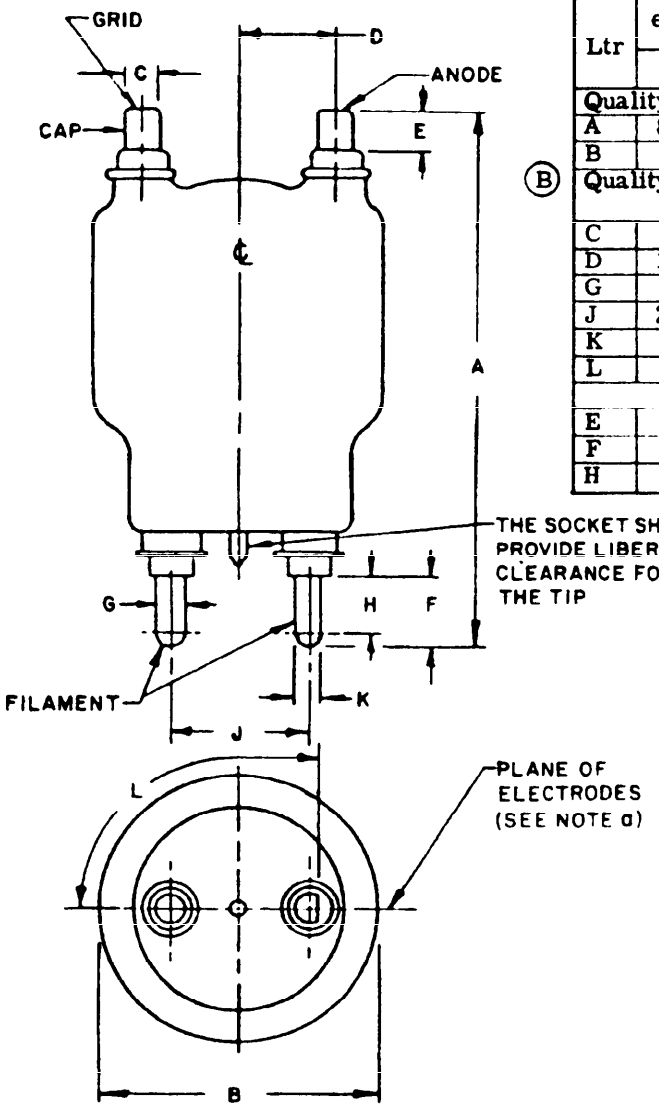
Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Ⓑ Preparing activity:
Air Force - 85

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	8.438 (214.33)	8.813 (223.85)
B		4.594 (116.69)
Ⓑ Quality conformance inspection, part 3 (see note b)		
C	.564 (14.33)	.570 (14.48)
D	1.469 (37.31)	1.531 (38.89)
G	.434 (11.02)	.440 (11.18)
J	2.062 (52.37)	2.188 (55.58)
K	.371 (9.42)	.379 (9.63)
L	90° ±7°	
Reference dimensions		
E	.625 (15.88)	
F	1.094 (27.79)	
H	.875 (22.23)	

NOTES:

a. The horizontal angle between the plane determined by the axes of the filament terminals and the plane determined by the axes of the grid and anode caps is not more than 5°.

b. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 833A.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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